may instead relate to the underlying cause of the pericardial effusion, such as fever or pleuritic chest pain in acute pericarditis or weight loss in cancer.

However, patients presenting with cardiac tamponade will appear ill and have signs and symptoms of **hemodynamic compromise**. Their blood pressure might be low and their jugular veins dilated, and they may also complain of dizziness and fatigue.

What Is the Pathophysiology of Pericardial Effusion and Cardiac Tamponade? (LO#2)

Pericardial effusions and cardiac tamponade are due to any process that leads to fluid from the bloodstream entering the pericardial space.

Causes of Pericardial Effusion

Pericardial effusions can be caused by almost any disease process that affects the pericardium:

- **Acute pericarditis** [eg, from viral infection, autoimmune disease, myocardial infarction, blunt/sharp trauma, uremia (symptomatic renal failure), ionizing radiation]—pericardial inflammation causes increased capillary permeability, resulting in effusion.
- **Nephrotic syndrome or liver failure**—decreased plasma oncotic pressure results in pericardial effusion as fluid moves from the blood into the interstitial space.
- **Malignancy**—usually caused by metastatic spread to the pericardium from other sites (malignant effusion).
- **Cardiac rupture** trating trauma) or aortic dissection extending to the pericardium—hemorrhage into the pericardial space results in effusion (hemorrhagic effusion or hemopericardium).
- **Hypothyroidism**—leading to increased capillary permeability and decreased lymphatic drainage.
- **Congestive Heart Failure**—elevated systemic venous pressures can lead to increased capillary pressures causing peripheral edema, pericardial effusions, and pleural effusions.
- **Idiopathic** In clinical practice, a substantial number of cases (> 25%) are ultimately labeled as idiopathic because a specific cause remains elusive despite comprehensive testing. Small, asymptomatic effusions found incidentally are often considered benign and are frequently idiopathic.

Pathophysiology of Cardiac Tamponade?

The causes of cardiac tamponade are the same as those of pericardial effusions. The effusion just becomes more severe, and the high volume compresses and compromises cardiac function—hence the term "tamponade" (compression).

Tamponade occurs when the accumulation of fluid within the pericardial cavity is large enough or rapid enough to increase the **pericardial pressure**, thus impeding the expansion and diastolic filling of the cardiac chambers. When the pressure inside the pericardium elevates, it will first compromise the low pressure right atrium (RA) and right ventricle (RV). Compression of the right heart chambers decreases venous return to the heart and therefore decreases left ventricle (LV) preload, cardiac output, and systemic blood pressure. Additionally, during inspiration there is still an increase in venous return, forcing more RV filling. This can push the mobile interventricular septum to the left, impairing LV g cardiac output and blood pressure (see pulsus paradoxus, below).

Compare the normal heart and the heart with cardiac tamponade in Figure 2. We can see how pericardial fluid accumulation can compress the cardiac chambers and limit cardiac output.

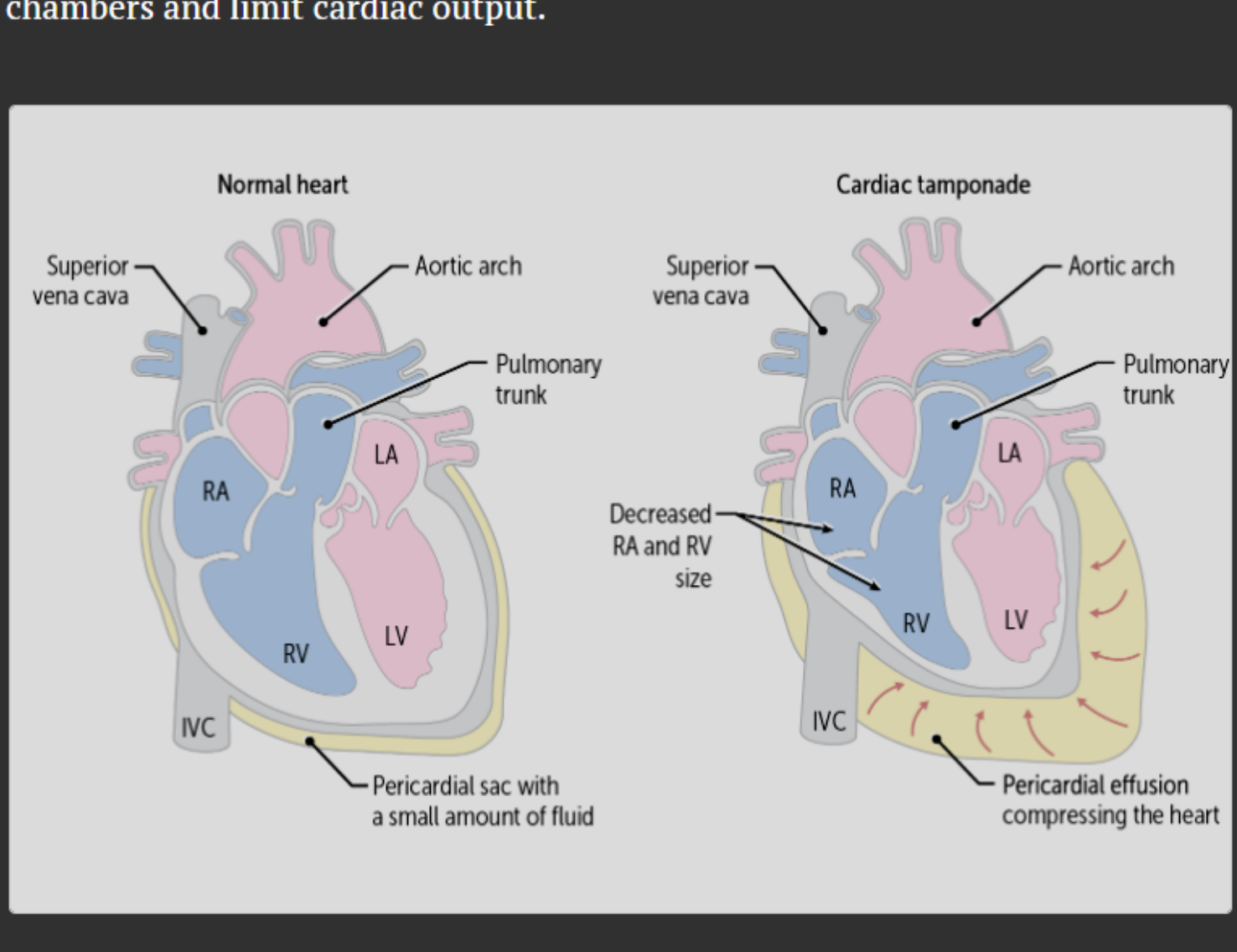


Figure 2. Pathophysiology of cardiac tamponade

The speed of fluid accumulation will determine much of the patient presentation. Slow, chronic fluid accumulation in the pericardium allows for the gradual stretchin more fluid, making urgent symptoms less common. On the other hand, rapid, acute accumulation of fluid (as in trauma-mediated hemopericardium) will cause more severe signs and symptoms because the fibrous pericardium cannot expand quickly - as little as 100 ml of blood in the pericardial sac can be lethal.

Compression of which cardiac chambers plays the biggest role in the development of complications in cardiac tamponade?

How Do We Diagnose Pericardial Effusion and Cardiac Tamponade? (LO#3)

When evaluating patients suspected of having a pericardial effusion, we should focus on answering three questions:

- Does the patient have a pericardial effusion?
- Is there hemodynamic compromise (ie, cardiac tamponade)?
- Can the specific on be identified?

We can answer these questions by combining the results of the history and physical exam with the results of electrocardiography (ECG), radiologic studies, and lab tests.

History

Small, chronic pericardial effusions are often discovered incidentally. In contrast, **rapidly developing pericardial effusions are high risk for tamponade** and hemodynamic compromise (eg, dizziness, dyspnea, or syncope due to low cardiac output). Other patient-history findings may help us identify the specific cause that led to the pericardial effusion (eg, malignancy, myocardial infarction, uremia). Chronic effusions can result in tamponade if they reach the limit of pericardial stretching.

What characteristic of a pericardial effusion is most important regarding the likelihood of developing symptoms?

Physical Examinati

a decrease in systolic blood pressure by >10 mm Hg during inspiration. Deep inspiration decreases intrathoracic pressure, which increases venous return. In tamponade, the low pressure right side of the heart is unable to expand, forcing the interventricular septum to the left. Left ventricular size decreases, decreasing stroke volume, cardiac output, and blood pressure.

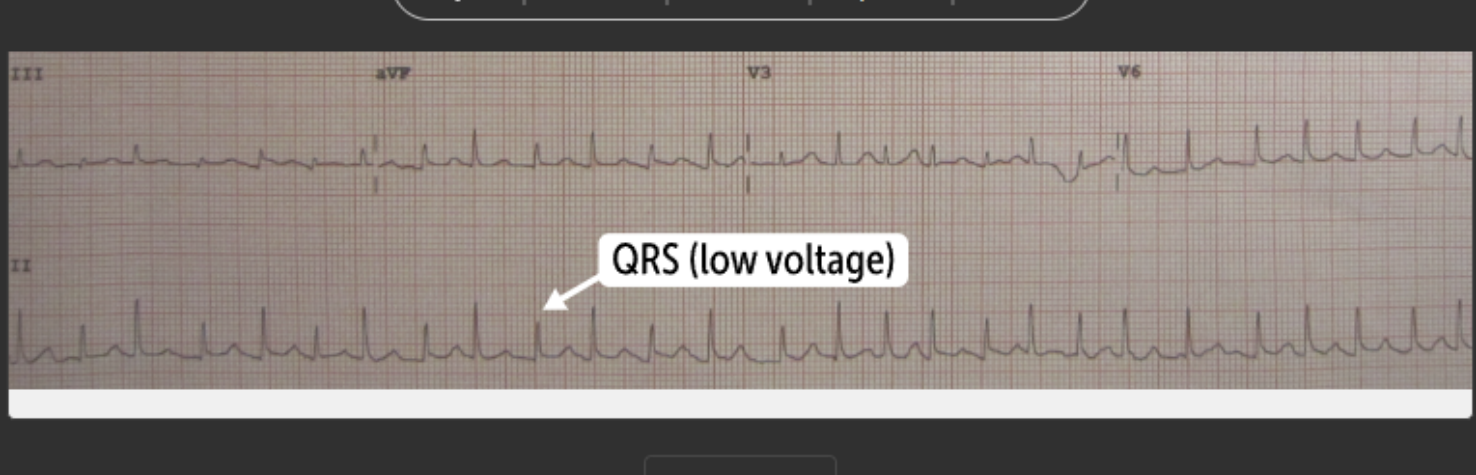
CLINICAL CORRELATION

Measuring pulsus paradoxus involves the following steps:

1. Inflate the blood pressure (BP) cuff until Korotkoff sounds are no longer heard.
2. Slowly deflate the BP cuff until sounds are heard intermittently (rather than with every heartbeat), and note the BP at this point. This corresponds to the higher systemic BP that occurs during expiration.
3. Continue to slowly deflate the cuff until sounds are heard with every beat, during inspiration and expiration, and note the BP at this point.
4. If the difference between the pressures noted during step 2 and step 3 is >10 mm Hg, this is considered pulsus paradoxus.

The original "paradox" meant that the radial pulse disappears during inspiration even though regular heart sounds are still auscultated with a stethoscope.

Notably, the Kussmaul sign (an increase in jugular venous pressure during inspiration) is usually not seen in cardiac tamponade and is instead a finding typical of constrictive



QUIZ

Figure 4. Electrical alternans

This sign is produced by the "swinging" movement of the heart in the fluid-filled pericardial sac. However, it's only seen in a minority of patients with tamponade (20%-25%).

ECG findings more commonly include sinus tachycardia and low-voltage QRS complexes in all leads. The latter occurs because the fluid around the heart limits electric conduction to the ECG leads. However, these findings are less specific.

Radiology Studies

A chest x-ray of a patient with pericardial effusion may reveal an enlarged heart silhouette with a "water bottle", rounded shape (Figure 5). Compare this to cardiomegaly, where the typical shape of the cardiac shadow is maintained. Note that to be detected by chest x-ray, at least 250 mL of fluid must be present in the pericardial cav noticed.

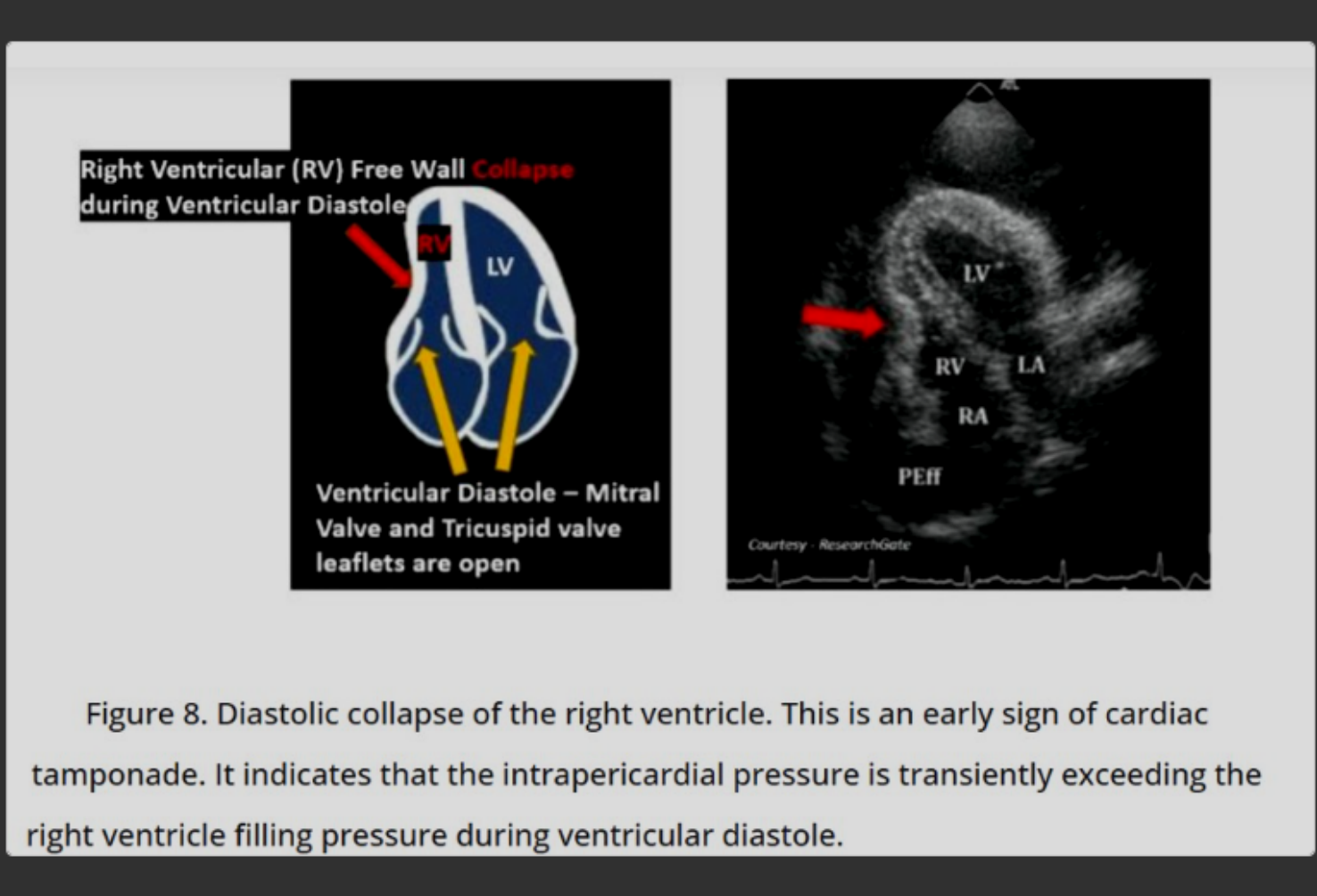
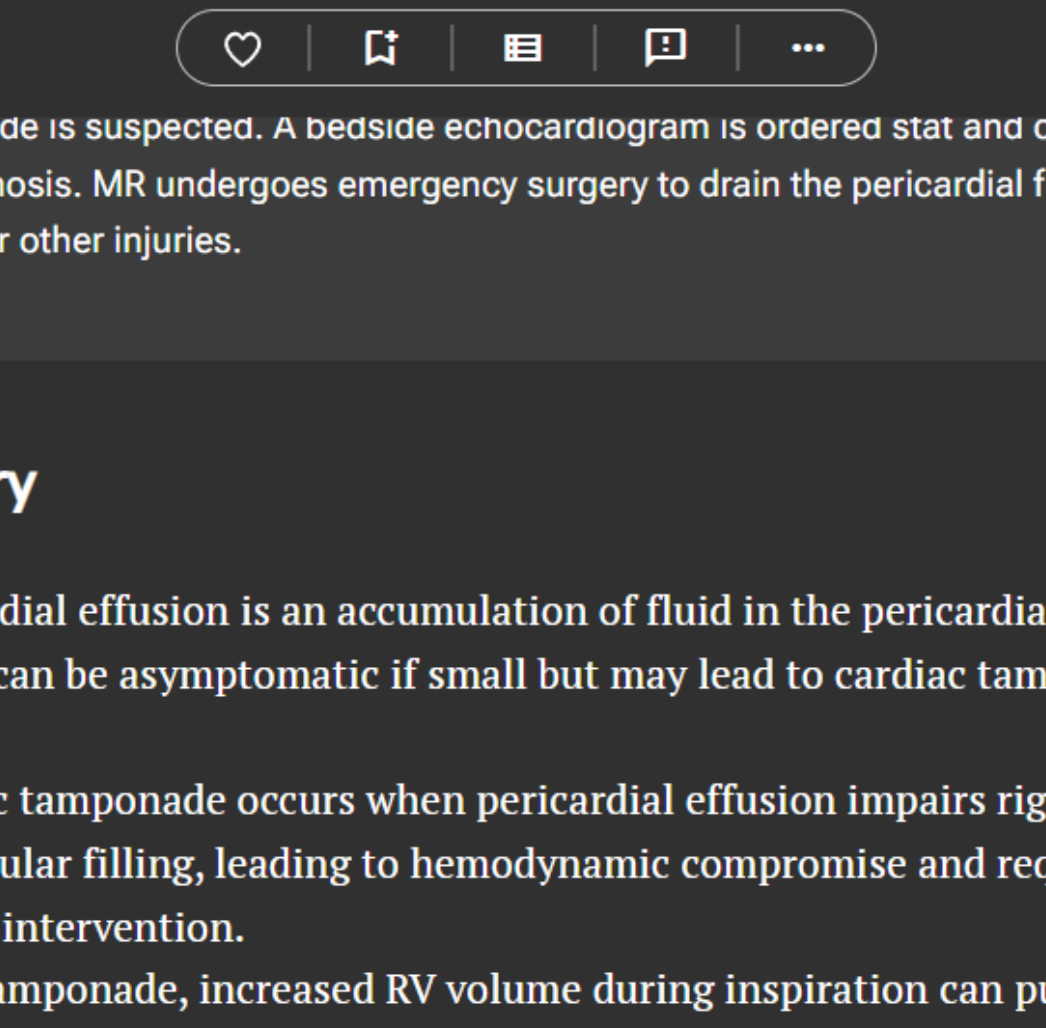

Right Ventricular (RV) Free Wall Collapse during Ventricular Diastole
RV
LV
Ventricular Diastole – Mitral Valve and Tricuspid valve leaflets are open
LV
RV
LA
RA
PEff
Courtesy · ResearchGate

Figure 8. Diastolic collapse of the right ventricle. This is an early sign of cardiac tamponade. It indicates that the intrapericardial pressure is transiently exceeding the right ventricle filling pressure during ventricular diastole.

Figure 6. Cardiac Tamponade on Echocardiography.

If a patient with pericardial effusion is hemodynamically stable with no evidence of cardiac tamponade, it is not necessary to drain the effusion. Treatment of the underlying condition may lead to clinical improvement or resolution without the need for drainage.

But in patients with cardiac tamponade and hemodynamic compromise, emergent drainage of the pericardial effusion is required. This is best achieved through **pericardiocentesis**, which involves the removal of fluid from the pericardial space via percutaneous (ie, through the skin) needle aspiration (Figure 7).



tamponade is suspected. A bedside echocardiogram is ordered stat and confirms the diagnosis. MR undergoes emergency surgery to drain the pericardial fluid and repair her other injuries.

Summary

- Pericardial effusion is an accumulation of fluid in the pericardial space, which can be asymptomatic if small but may lead to cardiac tamponade if large.
- Cardiac tamponade occurs when pericardial effusion impairs right ventricular filling, leading to hemodynamic compromise and requiring urgent intervention.
- With tamponade, increased RV volume during inspiration can push the interventricular septum towards the left ventricle, further decreasing cardiac output resulting in pulsus paradoxus.
- Causes of pericardial effusion include acute pericarditis, nephrotic syndrome, liver failure, malignancy, cardiac rupture, aortic dissection, and hypothyroidism.
- Beck triad (jugular venous distention, muffled heart sounds, and hypotension) is indicative of cardiac tamponade, along with pulsus paradoxus and electrical alternans on ECG.
- Echocardiography is the preferred imaging modality for diagnosing pericardial effusion and cardiac tamponade.
- Pericardiocentesis is the treatment of choice for cardiac tamponade, and a pericardial window may be used for continuous drainage in recurrent cases.

2. Which of the following is the most important factor in distinguishing cardiac tamponade from a nontamponade pericardial effusion?

A. ○ Flattened neck veins

B. ○ Displaced point of maximal impulse

C. ● Hypotension

D. ○ Hypertension

E. ○ Increased white blood cell count

Hide Explanation

The correct answer is hypotension, a sign of hemodynamic compromise characteristic of tamponade (C). As fluid continues to accumulate within the pericardial cavity, pressure in the pericardial space increases and compromises the filling of the right side of the heart. This reduces preload and left ventricle cardiac output. Cardiac tamponade can present with the Beck triad of distended (not flattened) neck veins (A), hypotension (not hypertension) (D), and muffled heart sounds. Neither a displaced point of maximal impulse (B) nor an increased white blood cell count are necessary for a diagnosis of cardiac tamponade (E).

3. A patient has established cardiac tamponade; what is the best next step in their managemen

B. ○ Myocardial infarction

C. ○ Hypothyroidism

D. ● Osteoporosis

E. ○ Uremia

Hide Explanation

The correct answer is osteoporosis (D). Cardiothoracic surgery (A), myocardial infarction, (B), hypothyroidism (C), and uremia (E) have all been associated with increased risk for pericardial effusion.

How would you rate this Brick?

References

- Appleton C, Gillam L, Koulogiannis K. Cardiac tamponade. *Cardiol Clin*. 2017;35(4):52